



Computational Chemistry Comparison and Benchmark DataBase

Release 22 (May

2022) Standard Reference Database 101 [National Institute of Standards and Technology](#)

Home	Experimental	Calculated	Comparisons
Resources	FAQ Help		

You are here: [Experimental](#) > [One molecule all properties](#)

Experimental data for C₃ (carbon trimer)

22 02 02 11 45

INChI	INChIKey	SMILES	IUPAC name
InChI=1S/C3/c1-3-2	NVLRFXKSQPKAD-UHFFFAOYSA-N	[C]=C=[C]	

State	Conformation
¹ Σ _g	D*H

Enthalpy of formation (H_{fg}), Entropy, Integrated heat capacity (0 K to 298.15 K) (HH), Heat Capacity (C_p)

Property	Value	Uncertainty	units	Reference	Comment
H _{fg} (298.15K) Δ _i H°	820.06		kJ mol ⁻¹	webbook	
H _{fg} (0K) Δ _i H°			kJ mol ⁻¹	webbook	
Entropy (298.15K) S°	237.27		J K ⁻¹ mol ⁻¹	webbook	
Integrated Heat Capacity (0 to 298.15K) H°-H°(T)	0.01		kJ mol ⁻¹	webbook	
Heat Capacity (298.15K) C _p °	37.74		J K ⁻¹ mol ⁻¹	webbook	

Information can also be found for this species in the [NIST Chemistry Webbook](#)

Vibrational levels (cm⁻¹)

Mode Number	Symmetry	Frequency			Intensity			Comment	Description
		Fundamental(cm ⁻¹)	Harmonic(cm ⁻¹)	Reference	(km mol ⁻¹)	unc.	Reference		
1	Σ _g	1225		webbook					
2	Σ _u	2040		webbook					
3	Π _u	63		webbook					

vibrational zero-point energy: 1695.7 cm⁻¹ (from fundamental vibrations)[Calculated vibrational frequencies](#) for C₃ (carbon trimer).

Rotational Constants (cm⁻¹)

See section [I.F.4 to change rotational constant units](#)

A	B	C	reference	comment
	0.43057		1998Ord/Say:2313	

[Calculated rotational constants](#) for C₃ (carbon trimer).

Product of moments of inertia I _A I _B I _C			
39.15168	amu Å ²	6.501391E-39	gm cm ²

Geometric Data

Point Group $D_{\infty h}$

Internal coordinates

distances (r) in Å, angles (a) in degrees, dihedrals (d) in degrees

Description	Value	unc.	Connectivity				Reference	Comment
			Atom 1	Atom 2	Atom 3	Atom 4		
rCC	1.277		1	2			1998Ord/Say:2313	
aCCC	180		2	1	3		1998Ord/Say:2313	

Cartesians

Atom	x (Å)	y (Å)	z (Å)
C1	0.0000	0.0000	0.0000
C2	0.0000	0.0000	1.2770
C3	0.0000	0.0000	-1.2770

Atom - Atom Distances

Distances in Å

	C1	C2	C3
C1		1.2770	1.2770
C2	1.2770		2.5540
C3	1.2770	2.5540	

[Calculated geometries](#) for C₃ (carbon trimer).

Experimental Bond Angles (degrees) from cartesians

atom1	atom2	atom3	angle
C2	C1	C3	180.000

Bond descriptions

Examples: C-C single bond, C=C, double bond, C#C triple bond, C:C aromatic bond

Bond Type	Count
C=C	2

Connectivity

Atom 1	Atom 2
C1	C2
C1	C3

Electronic energy levels (cm⁻¹)

Energy (cm ⁻¹)	Degeneracy	reference	description
0	1		$1\Sigma_g$
17090	1	webbook	

Ionization Energies (eV)

Ionization Energy	I.E. unc.	vertical I.E.	v.I.E. unc.	reference
13.000	0.100			webbook

Electron Affinity (eV)

Electron Affinity	unc.	reference
1.995	25.000	webbook

Dipole, Quadrupole and Polarizability**Electric dipole moment** $\leftrightarrow$

State	Config	State description	Conf description	Exp. min.	Dipole (Debye)				Reference	comment	Point Group	Components	
					x	y	z	total				dipole	quadrupole
1	1	$^1\Sigma_g$	$D_{\infty h}$	True	0.000	0.000	0.000	0.000			$D_{\infty h}$	0	1
1	2	1A_1	C_{2v}	False							C_{2v}	1	2

Experimental dipole measurement abbreviations: MW microwave; DT Dielectric with Temperature variation; DR Indirect (usually an upper limit); MB Molecular beam

[Calculated electric dipole moments](#) for C_3 (carbon trimer).

Electric quadrupole moment $+^-+$

State	Config	State description	Conf description	Exp. min.	Quadrupole (D Å)			Reference	comment	Point Group	Components	
					xx	yy	zz				dipole	quadrupole
1	1	$^1\Sigma_g$	$D_{\infty h}$	True						$D_{\infty h}$	0	1
1	2	1A_1	C_{2v}	False						C_{2v}	1	2

[Calculated electric quadrupole moments](#) for C_3 (carbon trimer).

References

By selecting the following links, you may be leaving NIST webspace. We have provided these links to other web sites because they may have information that would be of interest to you. No inferences should be drawn on account of other sites being referenced, or not, from this page. There may be other web sites that are more appropriate for your purpose. NIST does not necessarily endorse the views expressed, or concur with the facts presented on these sites. Further, NIST does not endorse any commercial products that may be mentioned on these sites. Please address comments about this page to cccbdb@nist.gov.

squib	reference	DOI
1998Ord/Say:2313	A Van Orden, RJ Saykally "Small Carbon Clusters, Spectroscopy, Structure and Energetics" Chem. Rev. 1998, 98, 2313-2357	10.1021/cr970086n
webbook	NIST Chemistry Webbook (http://webbook.nist.gov/chemistry)	10.18434/T4D303

Got a better number? Please email us at cccbdb@nist.gov

Browse	
Previous	Next
Li3+ (Lithium trimer cation)	C3+ (carbon trimer cation)